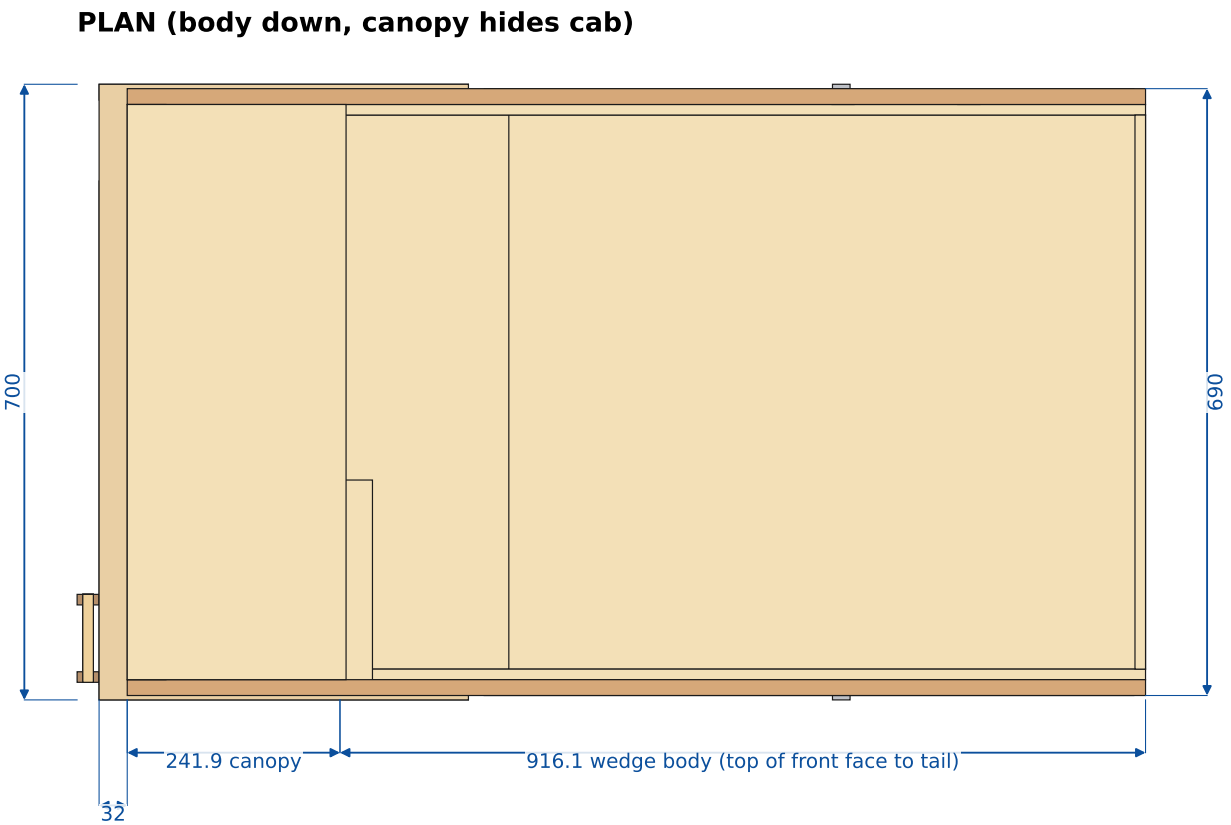
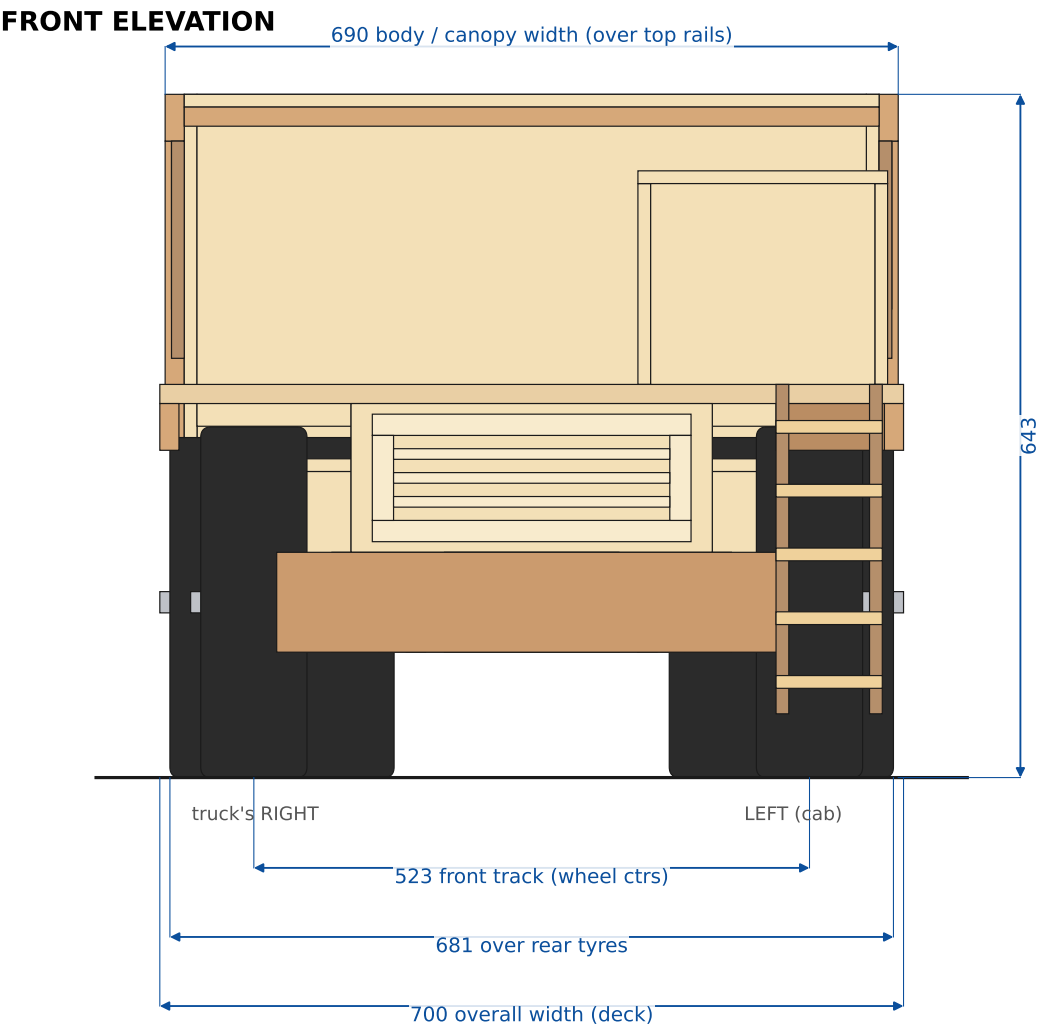
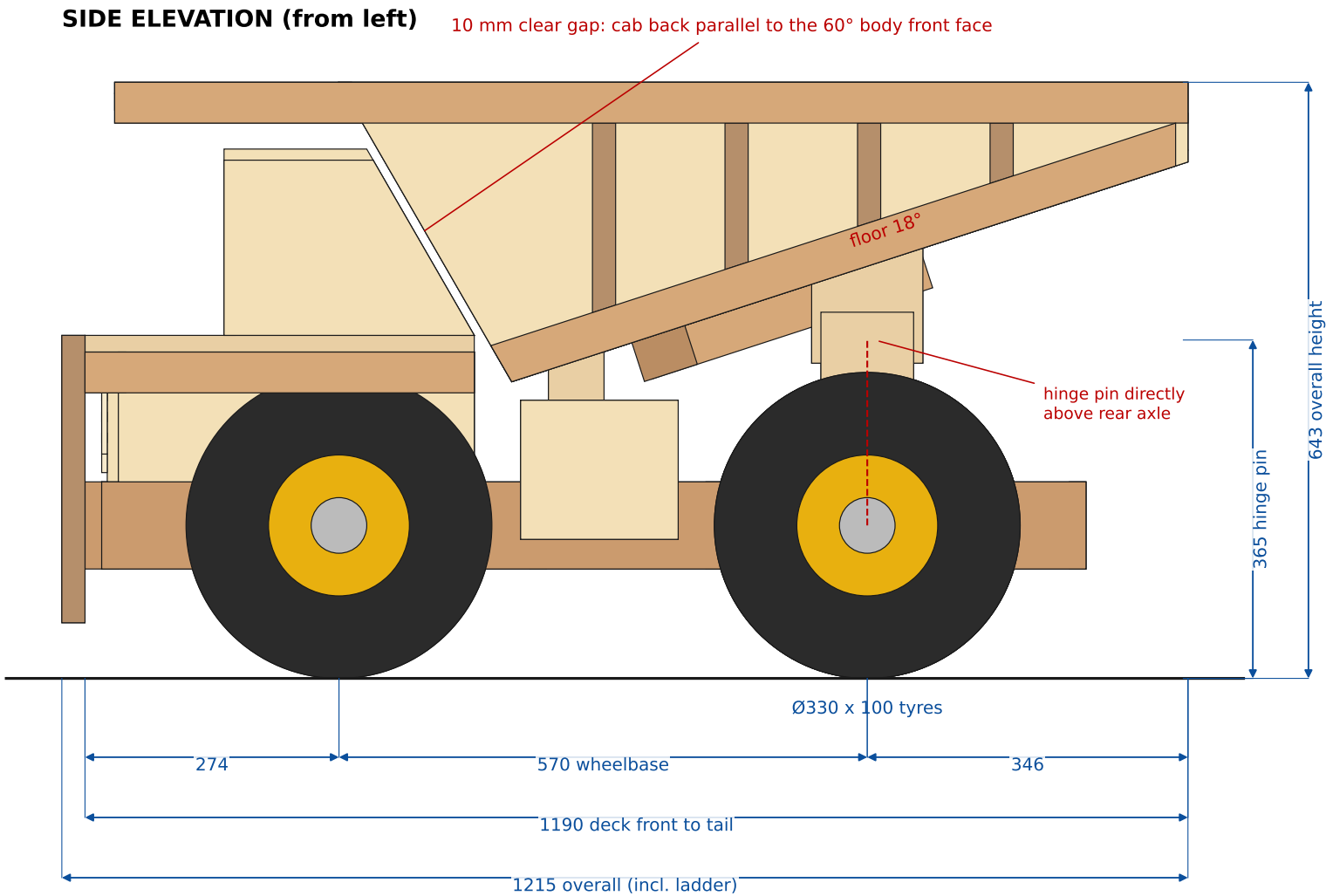
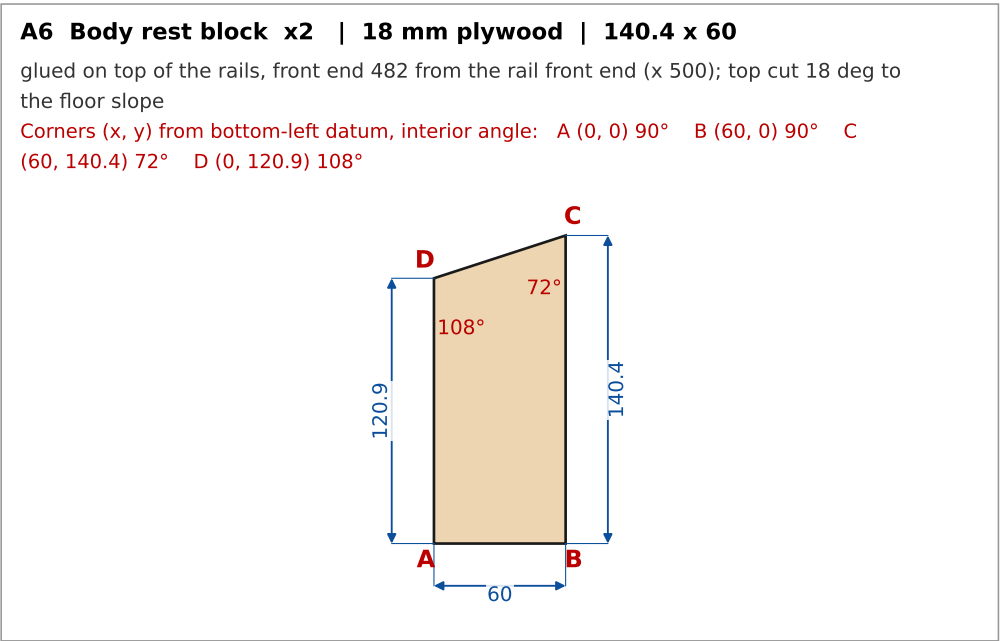
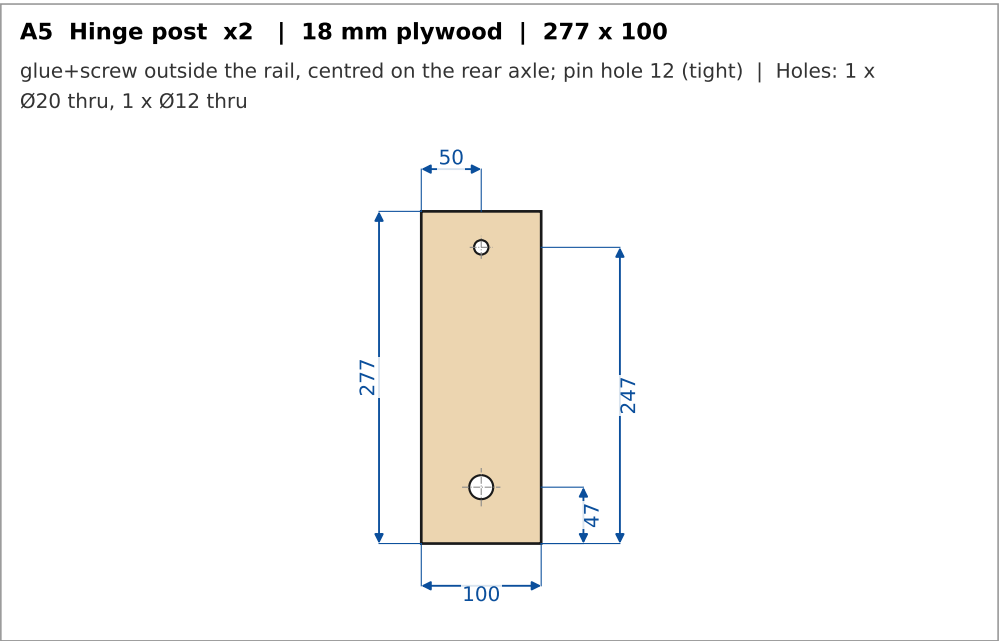
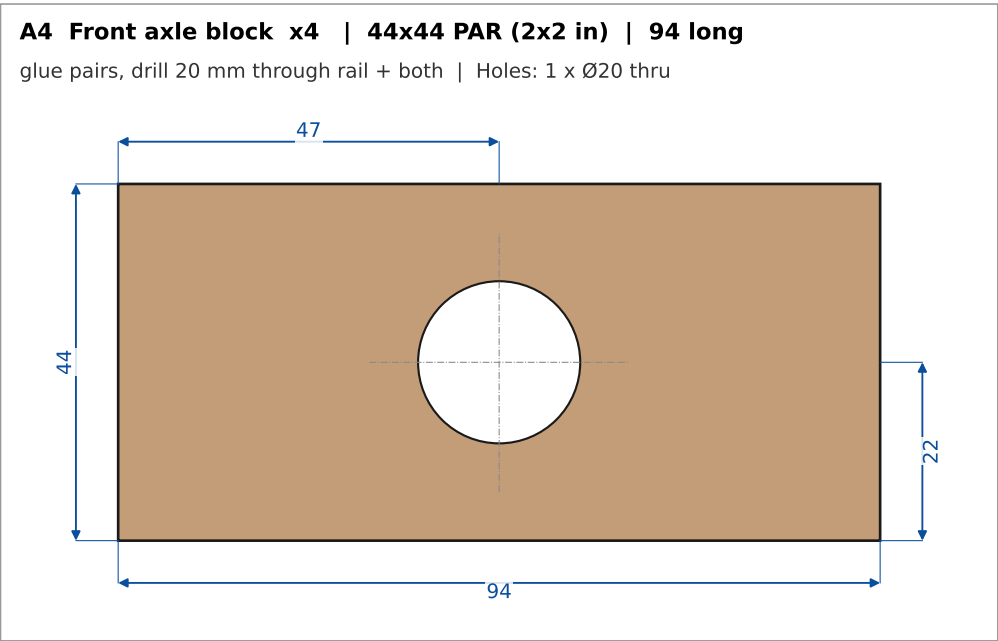
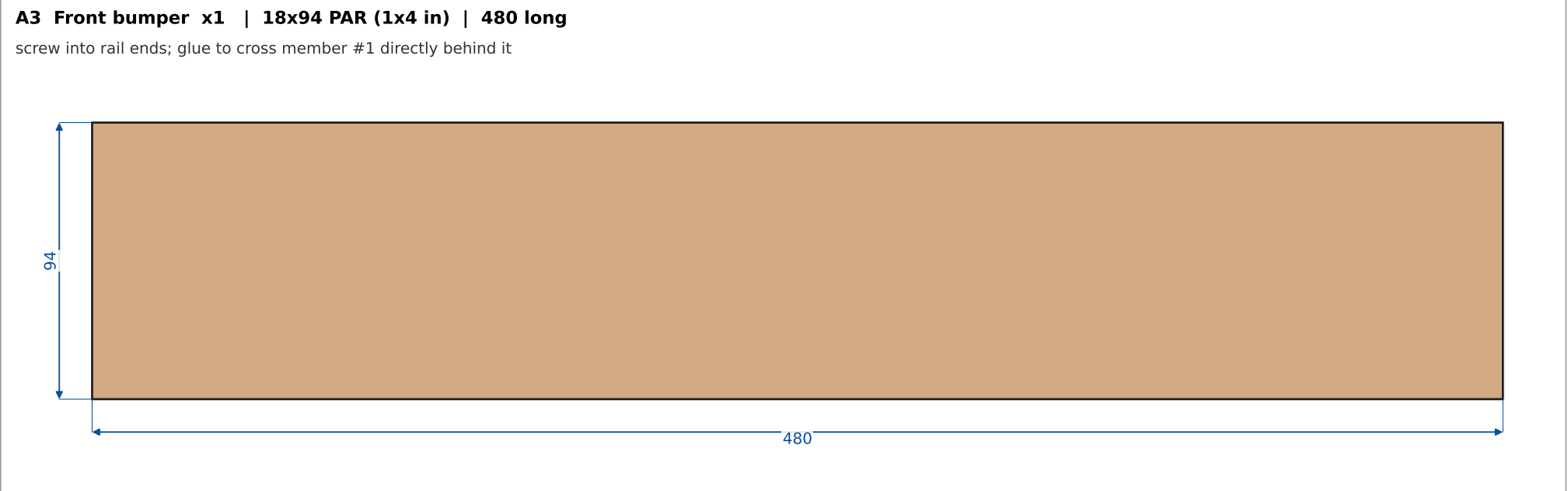
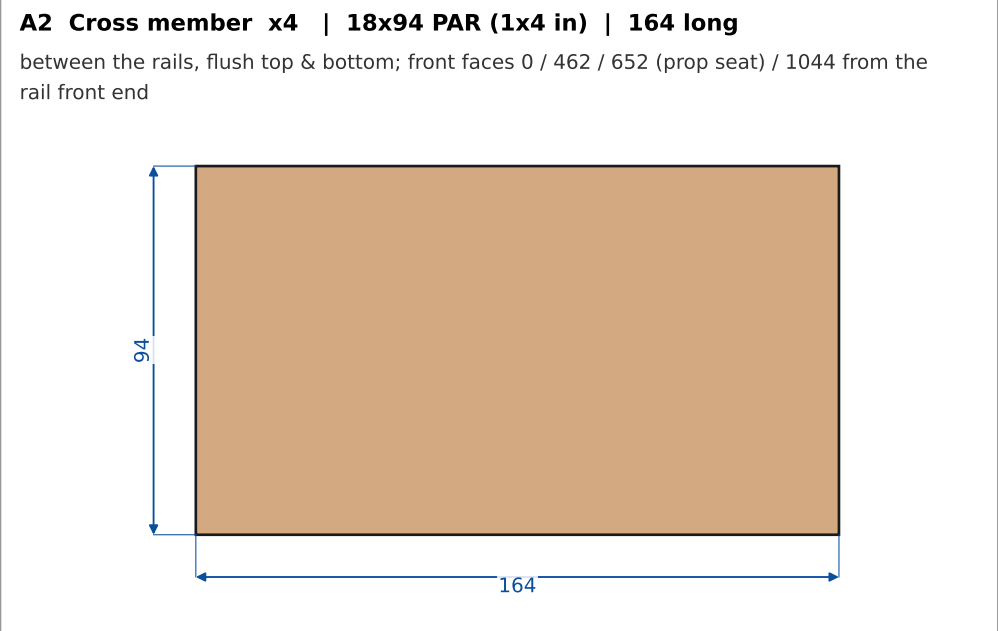
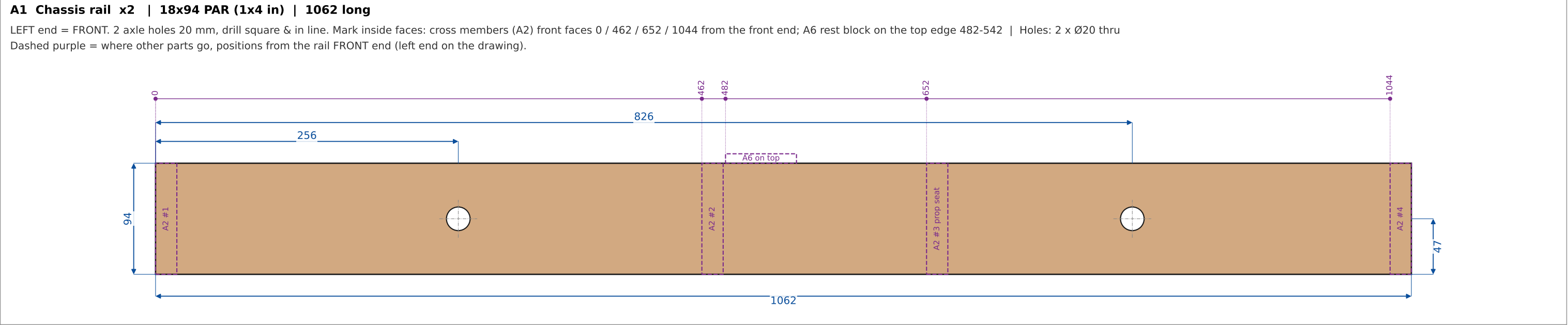


v3 WEDGE BODY. Side view from truck's LEFT (cab side), front view, plan view. Body shown down. Wheels: 4.00-6 PU puncture-proof 330 x 100 (purchased).



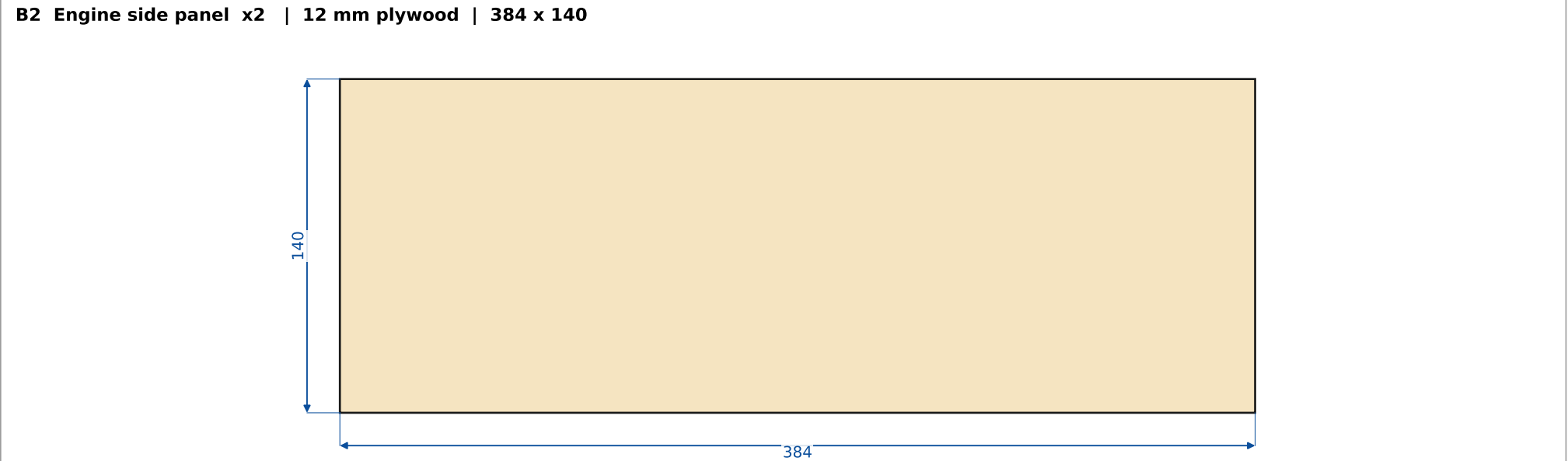
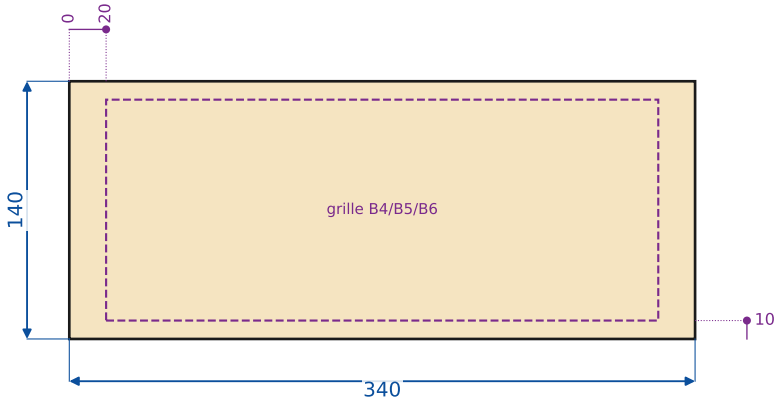
OVERALL	
Height (top of canopy)	643 mm (25.3 in)
Length (deck front - tail)	1190 mm; 1215 incl. ladder
Width (deck)	700 mm; body/canopy 690
Height, body tipped 45 deg	approx 1140 mm
WEDGE BODY	
Front face / cab back	60 deg from horizontal, parallel; clear gap 10 mm
Floor	rises 18 deg from the keel (z 320) to the tail (z 557.1)
Hinge	directly above rear axle (x 844), pin 365 above ground
Tip / tail clearance	45 deg; tail 256 mm clear of the ground
WHEELS / AXLES	
Wheel	4.00-6 PU puncture-proof, Ø330 x 100, hub 75, bore 20
Wheelbase / front track	570 mm / 523 mm (centres)
Rear duals: inner / outer ctrs	359 / 581 mm
Axles	Ø20 steel x 642 (front), x 700 (rear)
Tyre clearance (body down)	front: deck underside 22 above the tyre tops; body keel 77 mm clear of the tyres rear: floor underside 115 above the tyre tops (axle line); nearest: E9 hinge brackets, 31 mm from the inner tyres
SCALE CHECK vs CAT 777D (x1/8)	
Front canopy height	-> 643 (model 643)
Overall length	-> 1223 (model 1190)
Wheelbase	-> 571 (model 570)
Tyre 27.00R49 approx	-> 336 (model 330)
Body outside width	-> 690 (model 690)
Body shape	styled wedge, not to scale (Tony's sketch)

PAR = planed all round. Ply parts: length x width as cut. Hole centres and all positions measured from the bottom-left corner shown. Ø20 = axle holes (20 mm Forstner/auger, square to face). Bevelled panels: face view + edge profile.

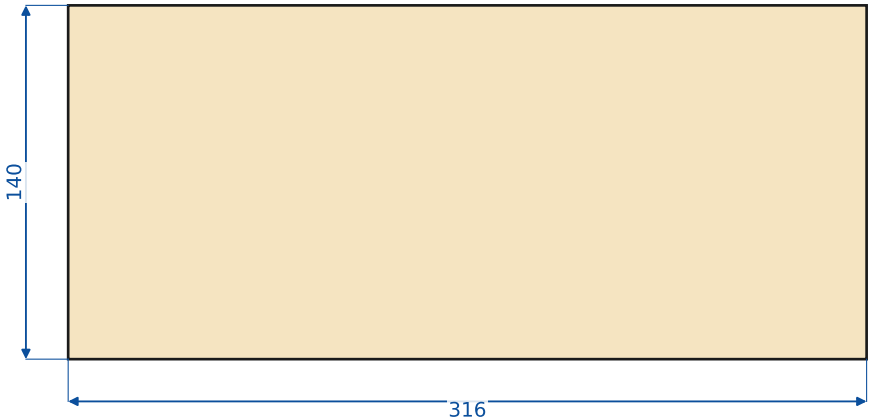


PAR = planed all round. Ply parts: length x width as cut. Hole centres and all positions measured from the bottom-left corner shown. Ø20 = axle holes (20 mm Forstner/auger, square to face). Bevelled panels: face view + edge profile.

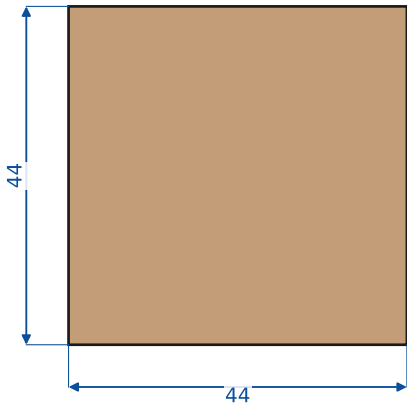
B1 Radiator front panel x1 | 12 mm plywood | 340 x 140
grille frame on the front face: 20 in from each side, 10 up from the bottom edge; B9 blocks glued behind the bottom corners
Dashed purple = where other parts go, positions front face.



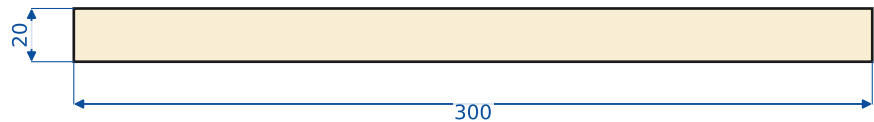
B3 Engine back panel x1 | 12 mm plywood | 316 x 140
between the sides; B9 blocks glued in front of the bottom corners



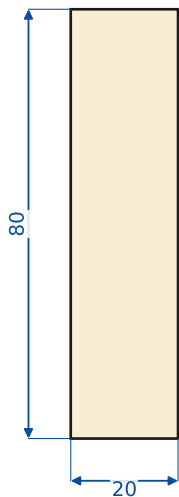
B9 Engine box hold-down block x4 | 44x44 PAR (2x2 in) | 44 long
inside the engine box, one on each rail behind B1 and in front of B3: glue to the panel, screw down into the rail top (4.0x70) 9 mm in from the block's inboard face, so it enters the middle of the 18 mm rail



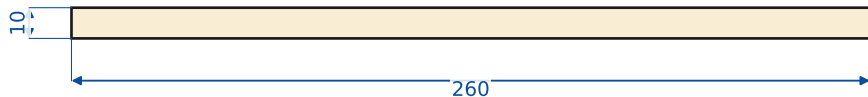
B4 Grille frame top/bottom x2 | 6 mm plywood | 300 x 20



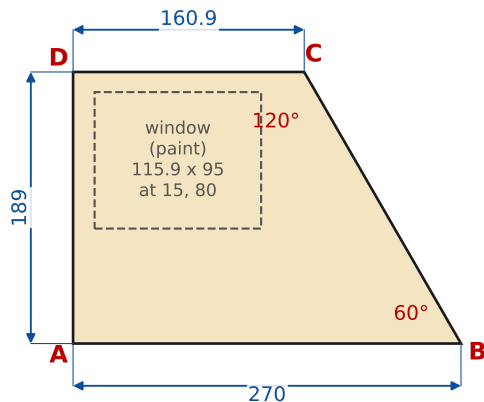
B5 Grille frame side x2 | 6 mm plywood | 80 x 20



B6 Grille slat x3 | 6 mm plywood | 260 x 10
12.5 mm gaps between slats



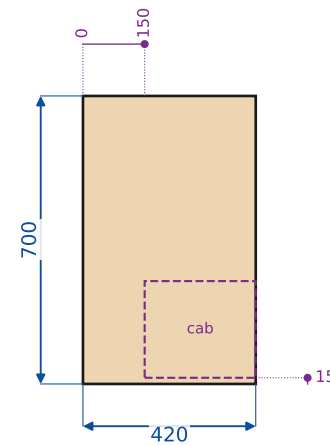
C1 Cab side x2 | 12 mm plywood | 270 x 189
back edge cut at 60 deg (parallel to body front); window = paint black, same position measured from the FRONT edge on both sides (mirror for the inner side)
Corners (x, y) from bottom-left datum, interior angle: A (0, 0) 90° B (270, 0) 60° C (160.9, 189) 120° D (0, 189) 90°



PAR = planed all round. Ply parts: length x width as cut. Hole centres and all positions measured from the bottom-left corner shown. Ø20 = axle holes (20 mm Forstner/auger, square to face). Bevelled panels: face view + edge profile.

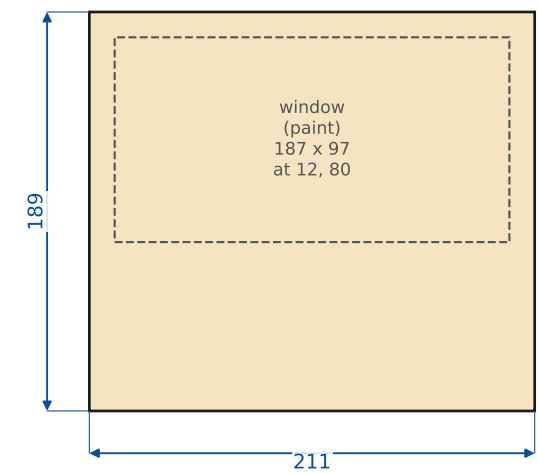
B7 Deck x1 | 18 mm plywood | 700 x 420

sits on the engine box. Cab on top: front 150 from the front edge, 15 in from the LEFT edge, cab back meets the rear edge. G3 ladder block glued underneath at the front-left corner
Dashed purple = where other parts go, positions seen from above: front edge on the left, truck's LEFT edge at the bottom.



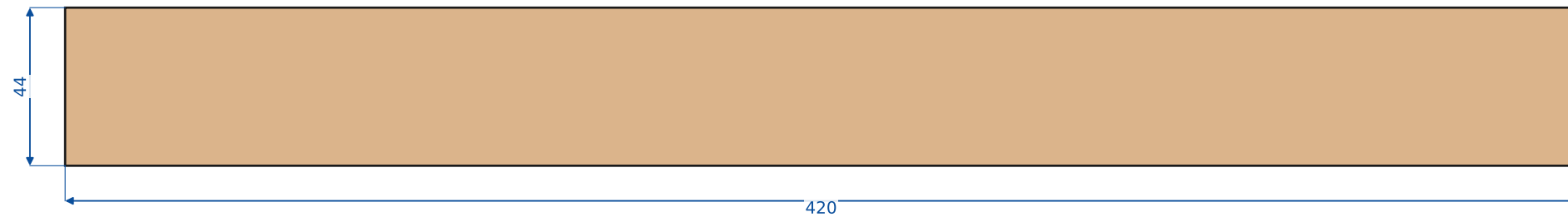
C2 Cab front x1	 12 mm plywood	 211 x 189
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windscreen = paint black (or cut)



B8 Deck edge rail x2 | 18x44 PAR (1x2 in) | 420 long

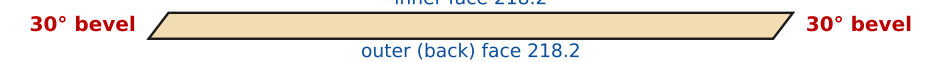
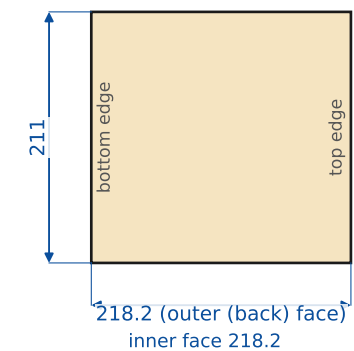
on edge under deck edges



C3 Cab back (sloping) x1 | 12 mm plywood | 225.2 x 211

at 60 deg between the sides, flush with their back edges. BOTH ends bevelled 30 deg, parallel, so it stands flat on the deck and the roof sits flat on it

Bevels: bottom edge bevelled 30° off square (long point on the outer (back) face); top edge bevelled 30° off square (long point on the inner face)

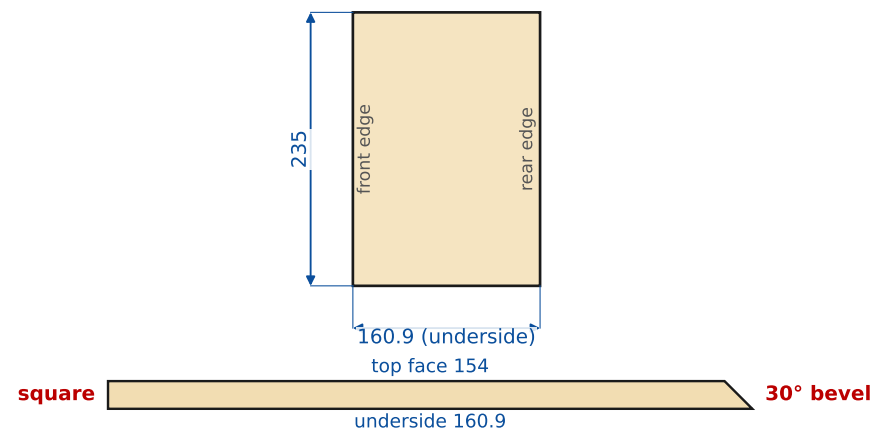


EDGE PROFILE (end view, thickness 12 exaggerated; cut from a 225.2 x 211 blank)

C4 Cab roof x1	 12 mm plywood	 235 x 160.9
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on the sides, flush front & sides; back edge bevelled 30 deg (underside longer) to continue the cab back slope

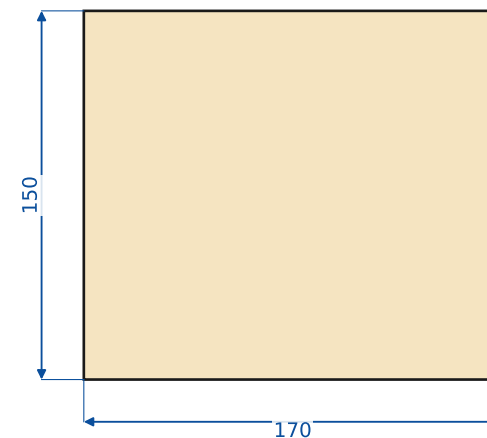
Bevels: front edge square; rear edge bevelled 30° off square (long point on the underside)



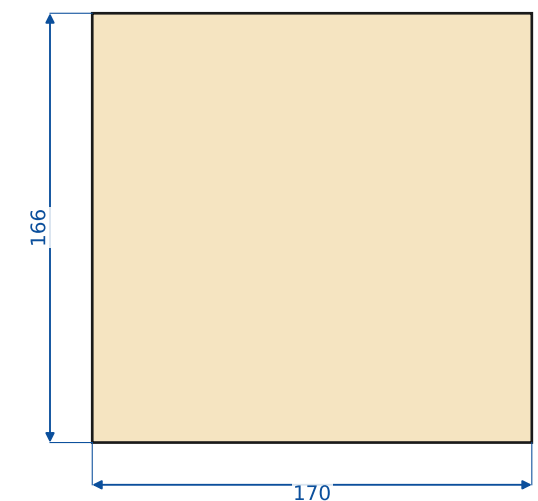
EDGE PROFILE (end view, thickness 12 exaggerated; cut from a 160.9 x 235 blank)

D1 Tank side (inner/outer) x4 | 12 mm plywood | 170 x 150

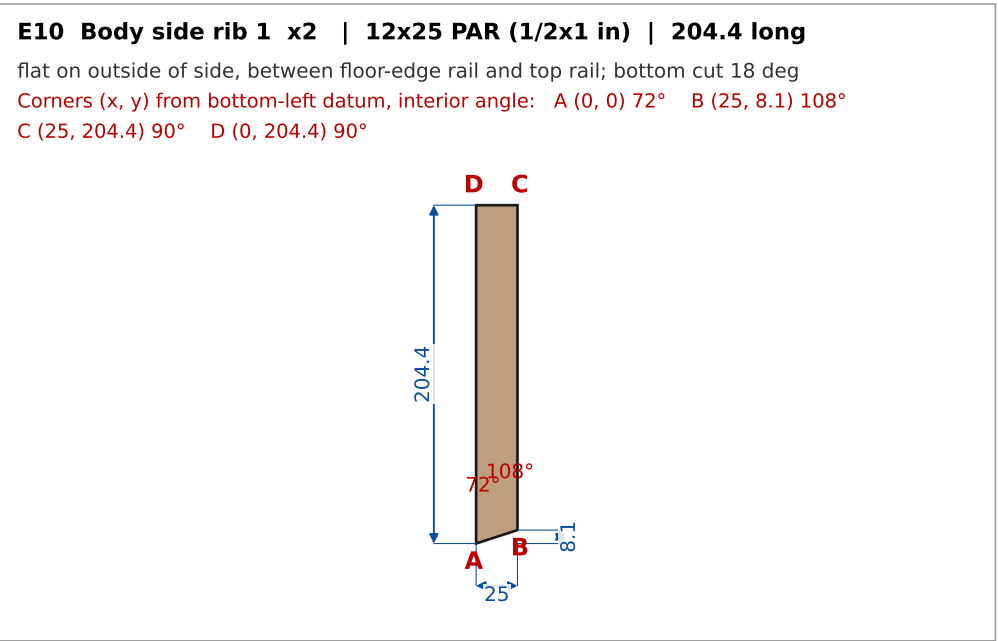
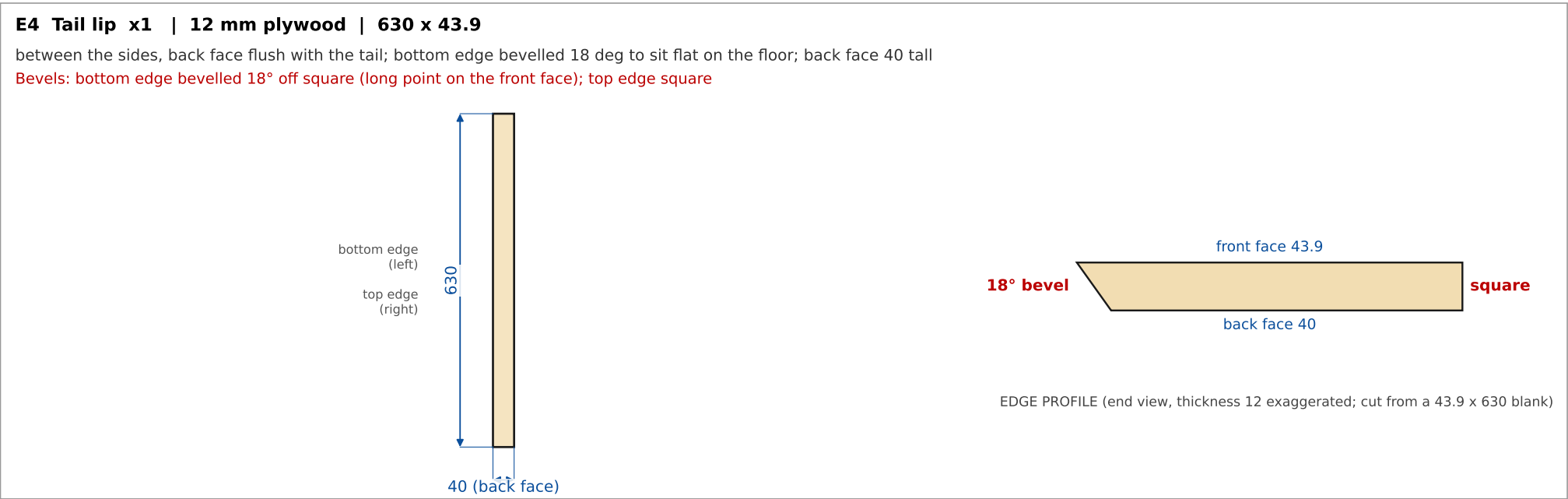
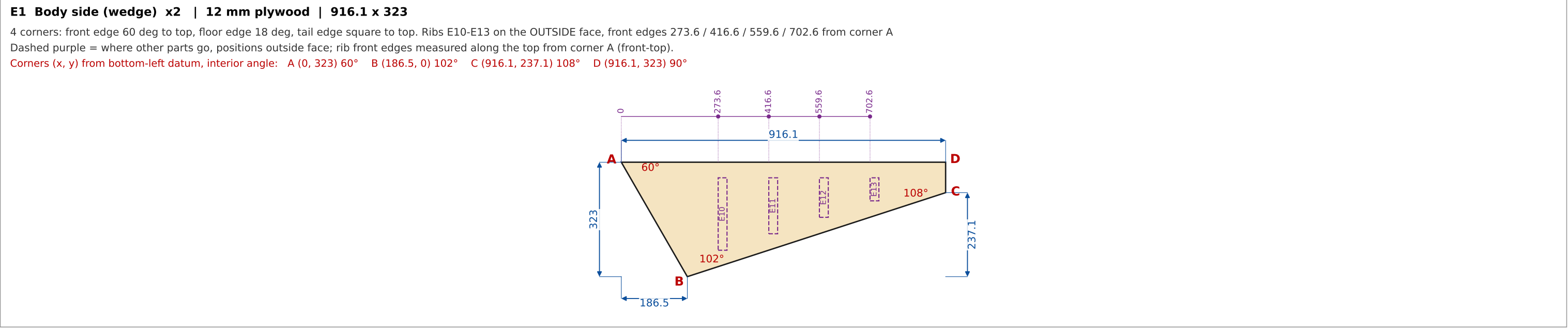
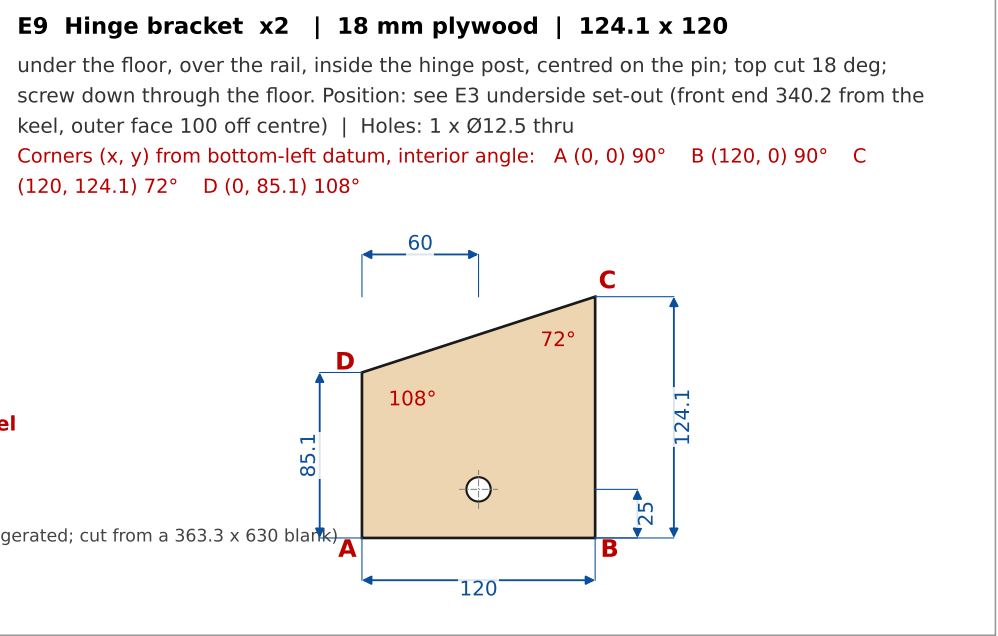
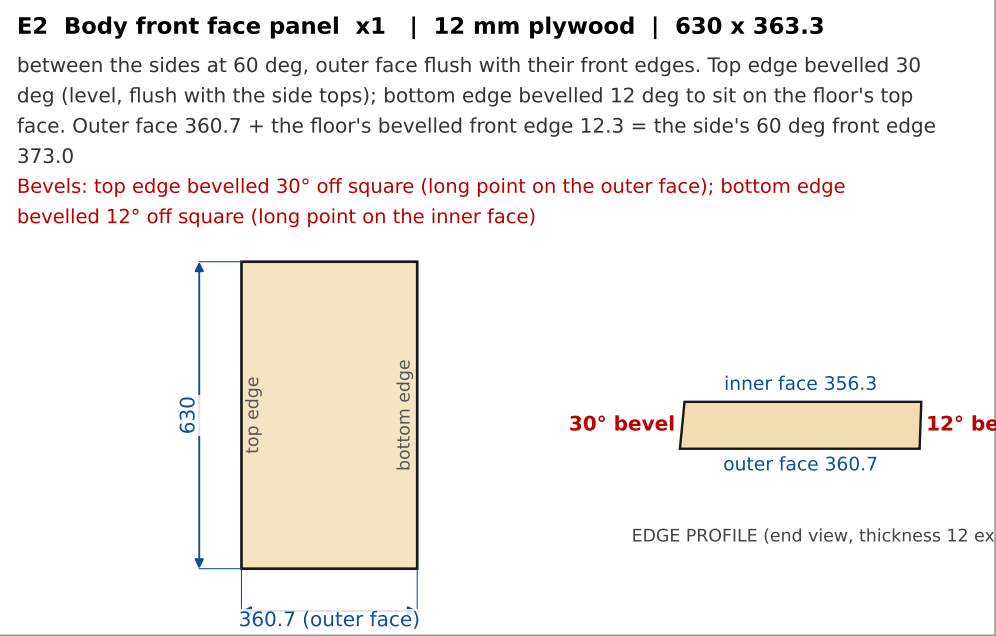
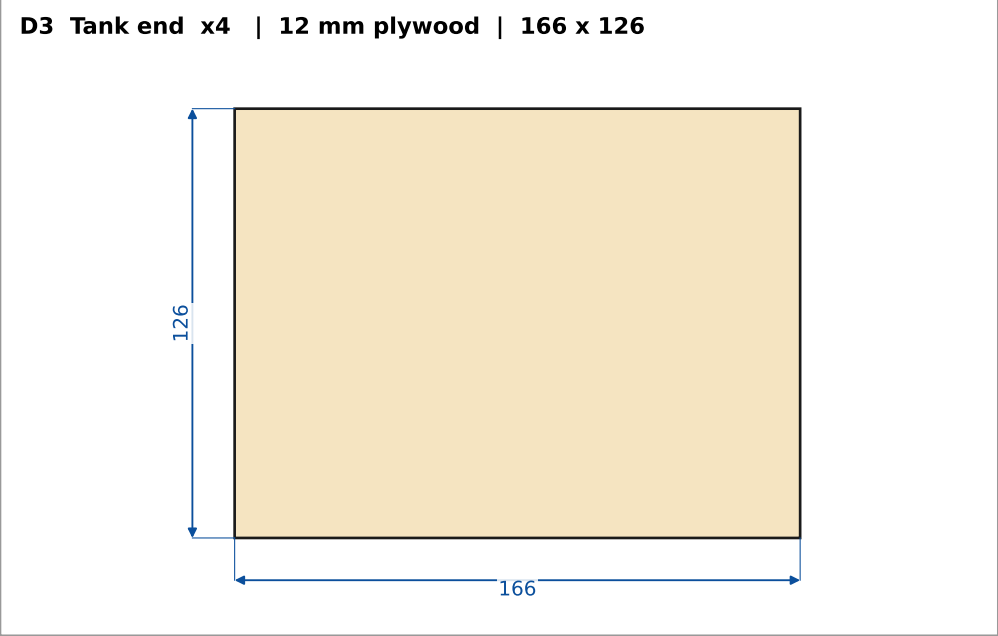
inner one screwed to the rail outer face first: front edge 452 from the rail front end (x 470), bottom edge 32 above the rail underside



D2 Tank top/bottom x4	 12 mm plywood	 170 x 166
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PAR = planed all round. Ply parts: length x width as cut. Hole centres and all positions measured from the bottom-left corner shown. Ø20 = axle holes (20 mm Forstner/auger, square to face). Bevelled panels: face view + edge profile.



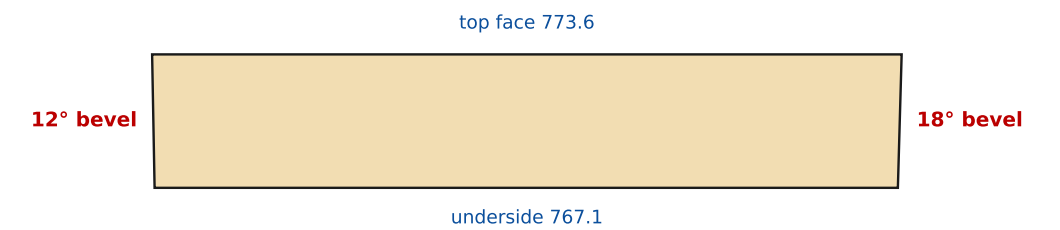
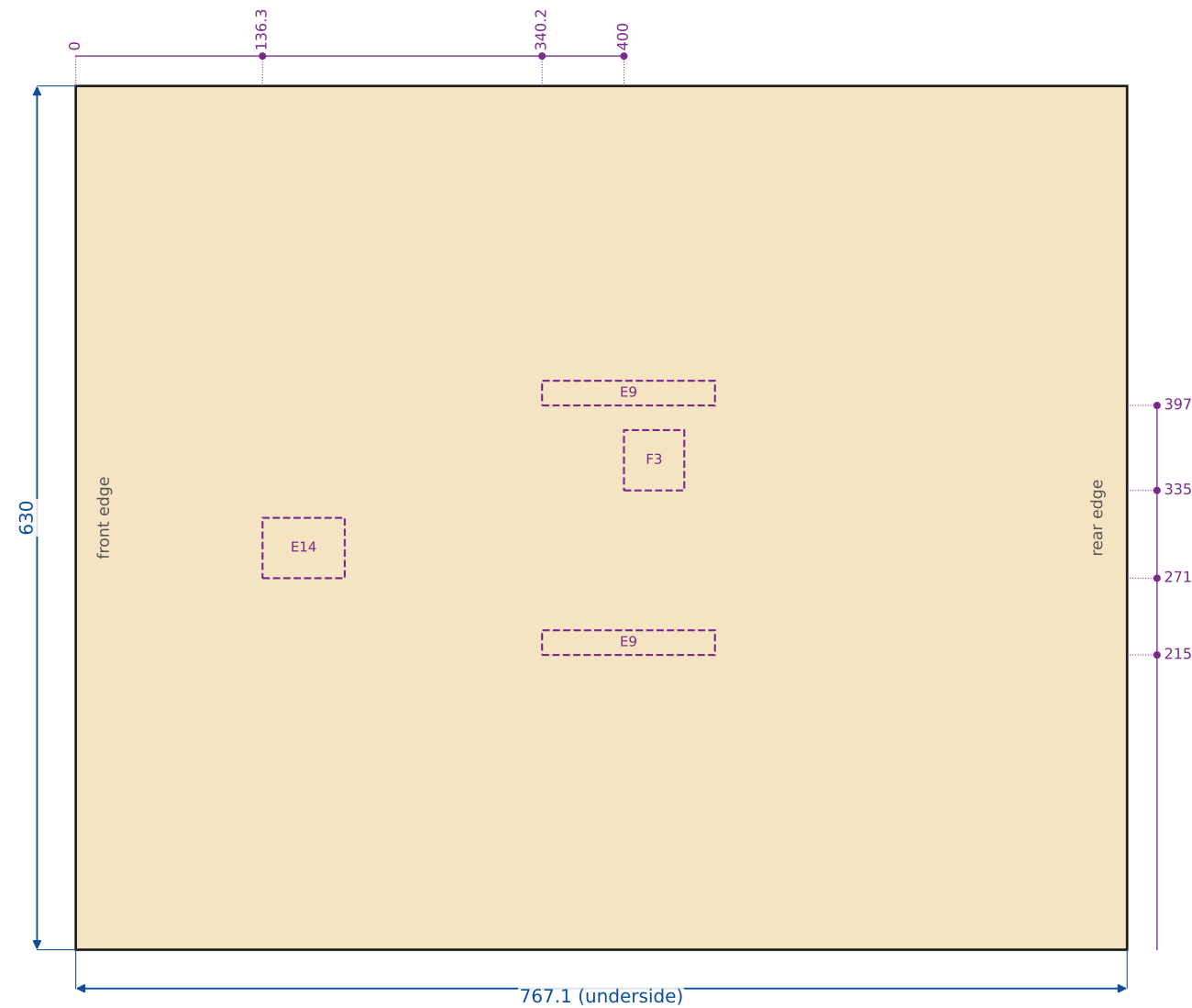
PAR = planed all round. Ply parts: length x width as cut. Hole centres and all positions measured from the bottom-left corner shown. Ø20 = axle holes (20 mm Forstner/auger, square to face). Bevelled panels: face view + edge profile.

E3 Body floor panel x1 | 12 mm plywood | 773.6 x 630

between the sides at 18 deg, underside flush with their bottom edges. Front edge bevelled 12 deg (on the 60 deg face line), rear edge 18 deg (plumb, flush with the tail). Underside set-out: E9 brackets 340.2-466.4 from the keel, 215-233 and 397-415 from the LEFT edge; E14 136.3-196.3, 271-315; F3 400-444, 335-379

Dashed purple = where other parts go, positions **UNDERSIDE**, measured from its front edge (the keel) and from its **LEFT** edge.

Bevels: front edge bevelled 12° off square (long point on the top face); rear edge bevelled 18° off square (long point on the top face)

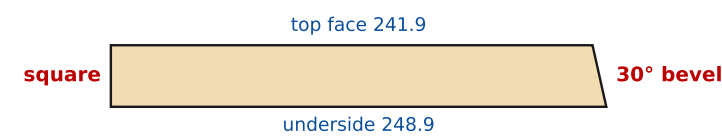
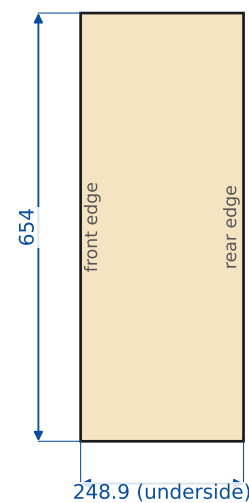


EDGE PROFILE (end view, thickness 12 exaggerated; cut from a 773.6 x 630 blank)

E5 Canopy x1 | 12 mm plywood | 654 x 248.9

between the top rails, flush with their tops and front ends; rear edge bevelled 30 deg (underside longer) to seat on the front face

Bevels: front edge square; rear edge bevelled 30° off square (long point on the underside)



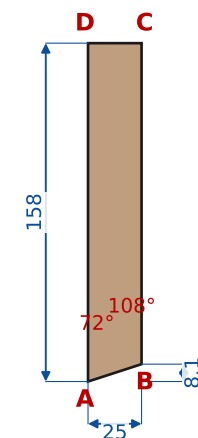
EDGE PROFILE (end view, thickness 12 exaggerated; cut from a 248.9 x 654 blank)

E11 Body side rib 2 x2 | 12x25 PAR (1/2x1 in) | 158 long

flat on outside of side, between floor-edge rail and top rail; bottom cut 18 deg

Corners (x, y) from bottom-left datum, interior angle: A (0, 0) 72° B (25, 8.1) 108°

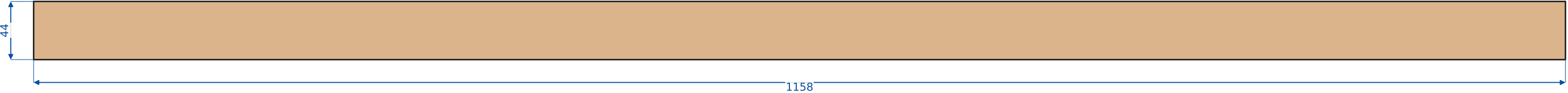
C (25, 158) 90° D (0, 158) 90°



PAR = planed all round. Ply parts: length x width as cut. Hole centres and all positions measured from the bottom-left corner shown. Ø20 = axle holes (20 mm Forstner/auger, square to face). Bevelled panels: face view + edge profile.

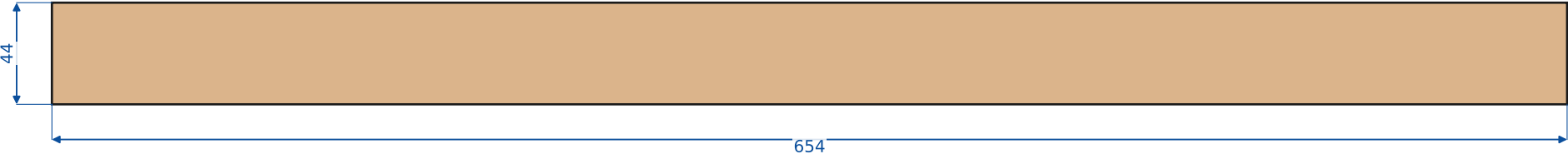
E6 Body top rail x2 | 18x44 PAR (1x2 in) | 1158 long

flat on outside of side, flush with top; runs 241.9 ahead of corner A to carry the canopy edge



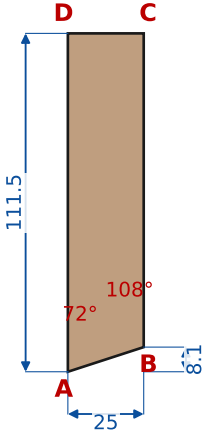
E7 Canopy front stiffener x1 | 18x44 PAR (1x2 in) | 654 long

laid FLAT under the canopy front edge, flush with its front, between the top rails (hidden behind them)



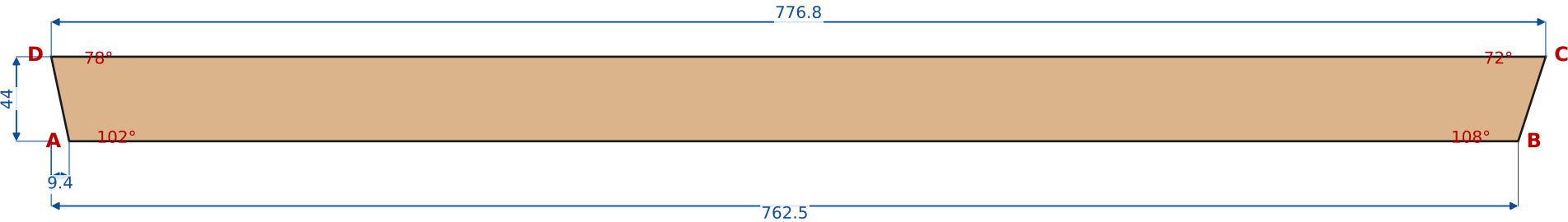
E12 Body side rib 3 x2 | 12x25 PAR (1/2x1 in) | 111.5 long

flat on outside of side, between floor-edge rail and top rail; bottom cut 18 deg
Corners (x, y) from bottom-left datum, interior angle: A (0, 0) 72° B (25, 8.1) 108°
C (25, 111.5) 90° D (0, 111.5) 90°



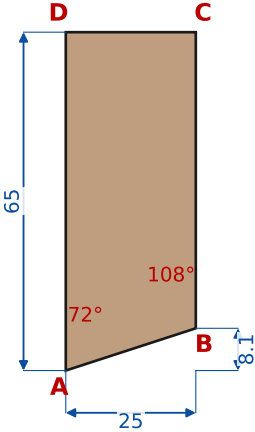
E8 Floor-edge rail x2 | 18x44 PAR (1x2 in) | 776.8 long

two identical pieces (flip one). Flat on outside of side along the floor edge, front end on the 60 deg edge, rear end plumb; it stops 13.3 short of the tail where its top edge meets the top rail. Ends cut 78 deg (front) and 72 deg (rear) to the long edge
Corners (x, y) from bottom-left datum, interior angle: A (9.4, 0) 102° B (762.5, 0) 108° C (776.8, 44) 72° D (0, 44) 78°



E13 Body side rib 4 x2 | 12x25 PAR (1/2x1 in) | 65 long

flat on outside of side, between floor-edge rail and top rail; bottom cut 18 deg
Corners (x, y) from bottom-left datum, interior angle: A (0, 0) 72° B (25, 8.1) 108°
C (25, 65) 90° D (0, 65) 90°



PAR = planed all round. Ply parts: length x width as cut. Hole centres and all positions measured from the bottom-left corner shown. Ø20 = axle holes (20 mm Forstner/auger, square to face). Bevelled panels: face view + edge profile.

E14 Prop pivot block x1 | 44x44 PAR (2x2 in) | 60 long

glued+screwed under the floor, RIGHT face on the centreline, front end 136.3 from the keel (see E3); the prop pivots on its RIGHT side | Holes: 1 x Ø6.5 thru

F1 Prop arm x1 | 18x44 PAR (1x2 in) | 319 long

M6 bolt pivot; folds under the floor and is held by the F4 turn-button; stands on the 3rd cross member at 45 deg | Holes: 1 x Ø6.5 thru

F3 Prop catch block x1 | 44x44 PAR (2x2 in) | 44 long

glued+screwed under the floor 2 mm to the RIGHT of the folded prop, front end 400 from the keel (see E3)

F4 Prop turn-button x1 | 12x25 PAR (1/2x1 in) | 70 long

screwed to the underside of F3 (4.0x50 + washer, snug but turnable); turn across under the prop to hold it folded, turn along the floor to release | Holes: 1 x Ø5 thru

G1 Ladder stile x2 | 12x25 PAR (1/2x1 in) | 310 long

12x25 on edge (25 deep front-to-back). The ladder is 100 wide overall: its two outside faces are 330 and 230 from the centreline on the LEFT, tops flush with the deck; each stile screwed (4.0x40) to the deck edge and the G3 block | Holes: 5 x Ø12 thru

G3 Ladder fixing block x1 | 44x44 PAR (2x2 in) | 102 long

glued under the deck at the front-left corner, against the deck edge rail (B8), front face flush with the deck front

H3 Spacer washer 18 mm x8 | 18 mm plywood | Ø60 disc

4 front (inboard of wheel), 4 between rear duals | Holes: 1 x Ø20 thru

H4 Spacer washer 12 mm x4 | 12 mm plywood | Ø60 disc

between hinge post and inner rear wheel | Holes: 1 x Ø20 thru

H1 Front axle x1 | 20 mm bright mild steel round bar | 642 long

held by collars (does not turn; wheels turn on it); or drill 4 mm split-pin holes 17 mm in from each end

PAR = planed all round. Ply parts: length x width as cut. Hole centres and all positions measured from the bottom-left corner shown. Ø20 = axle holes (20 mm Forstner/auger, square to face). Bevelled panels: face view + edge profile.

H2 Rear axle x1 | 20 mm bright mild steel round bar | 700 long

held by collars (does not turn; wheels turn on it)

Ø20 20 mm bright mild steel round bar

700

F2 Body hinge pin (stub) x2 | 12 mm hardwood dowel | 42 long

glue into hinge post only (12 mm hole); body bracket hole 12.5 so it swings

Ø12 12 mm hardwood dowel

42

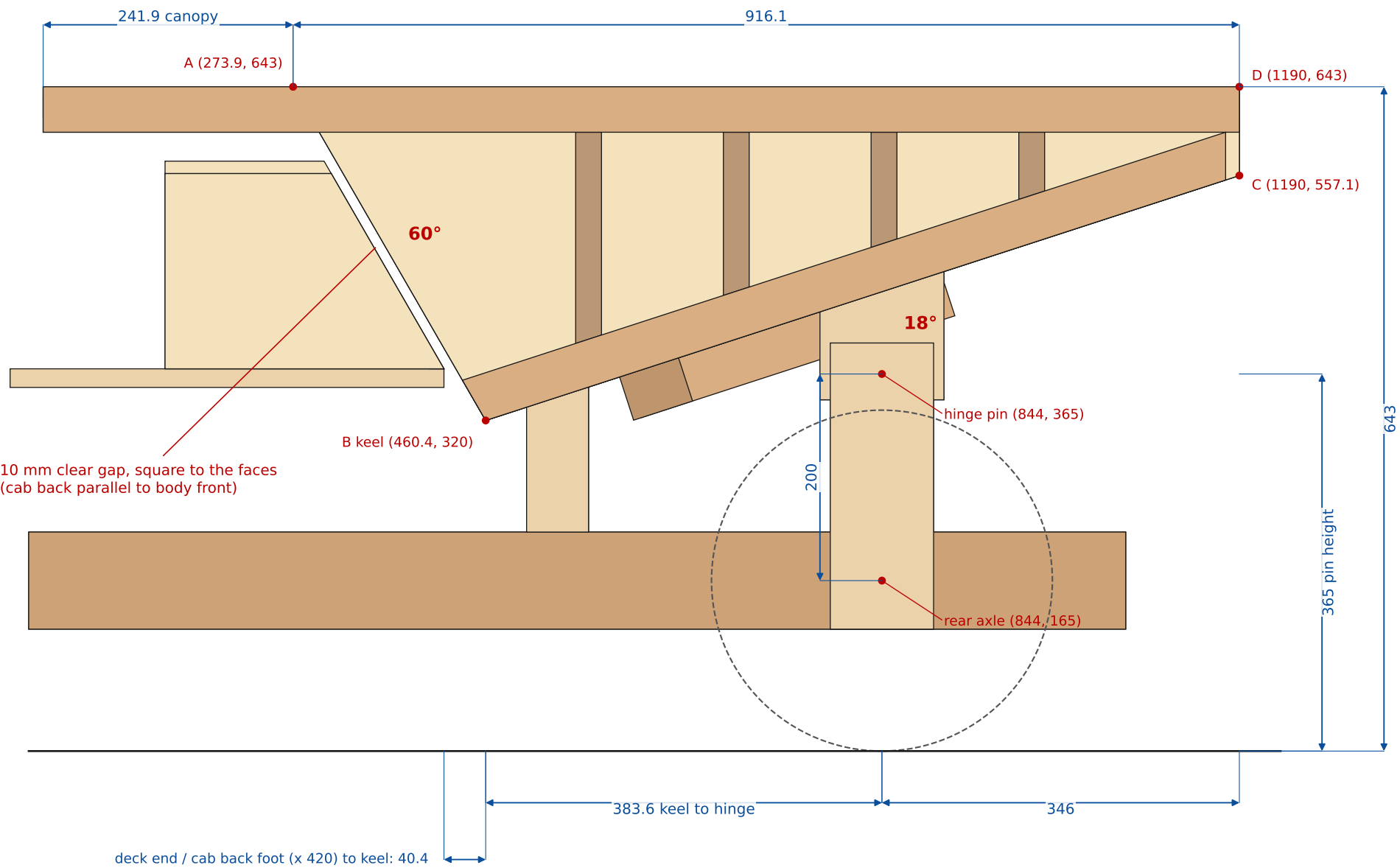
G2 Ladder rung x5 | 12 mm hardwood dowel | 100 long

glue into stile holes

Ø12 12 mm hardwood dowel

100

WEDGE BODY + HINGE SET-OUT (side view from left, body down; x from deck front, z from ground)



Hinge brackets E9 sit inside the hinge posts A5, over the rails; the prop F1 lies folded under the floor, held by the F4 turn-button on the F3 block.

Lengths/sizes are finished sizes in mm. PAR sizes are ACTUAL planed sizes (nominal inch size in brackets). Cut long parts first; check against real stock before cutting.

Part	Name	Material	Qty	Length (PAR) / L x W (ply)	Thk / Ø	Notes
A1	Chassis rail	18x94 PAR (1x4)	2	1062	18	LEFT end = front; 2 axle holes; mark cross-member + rest-block positions (drawing); holes 2 x Ø20 (see drawing)
A2	Cross member	18x94 PAR (1x4)	4	164	18	front faces 0 / 462 / 652 (prop seat) / 1044 from the rail front end
A3	Front bumper	18x94 PAR (1x4)	1	480	18	screw into rail ends; glue to cross member #1 directly behind it
A4	Front axle block	44x44 PAR (2x2)	4	94	44	glue pairs, drill 20 mm through rail + both; holes 1 x Ø20 (see drawing)
A5	Hinge post	18 ply	2	277 x 100	18	glue+screw outside the rail, centred on the rear axle; pin hole 12 (tight); holes 1 x Ø20, 1 x Ø12 (see drawing)
A6	Body rest block	18 ply	2	140.4 x 60	18	top cut 18 deg; front end 482 from the rail front end
B1	Radiator front panel	12 ply	1	340 x 140	12	grille 20 in from each side, 10 up from the bottom
B2	Engine side panel	12 ply	2	384 x 140	12	
B3	Engine back panel	12 ply	1	316 x 140	12	between the sides
B9	Engine box hold-down block	44x44 PAR (2x2)	4	44	44	inside the engine box on the rails; glue to B1/B3, screw down (4.0x70)
B4	Grille frame top/bottom	6 ply	2	300 x 20	6	
B5	Grille frame side	6 ply	2	80 x 20	6	
B6	Grille slat	6 ply	3	260 x 10	6	12.5 mm gaps between slats
B7	Deck	18 ply	1	700 x 420	18	cab 150 back, 15 in from the LEFT edge; G3 underneath front-left
B8	Deck edge rail	18x44 PAR (1x2)	2	420	18	on edge under deck edges
C1	Cab side	12 ply	2	270 x 189	12	back edge 60 deg; window paint, measured from the front edge
C2	Cab front	12 ply	1	211 x 189	12	windscreen = paint black (or cut)
C3	Cab back (sloping)	12 ply	1	225.2 x 211	12	both ends bevelled 30 deg, parallel (see edge profile)
C4	Cab roof	12 ply	1	235 x 160.9	12	back edge bevelled 30 deg (see edge profile)
D1	Tank side (inner/outer)	12 ply	4	170 x 150	12	inner one on the rail: 452 from the rail front end, 32 above its underside
D2	Tank top/bottom	12 ply	4	170 x 166	12	
D3	Tank end	12 ply	4	166 x 126	12	
E1	Body side (wedge)	12 ply	2	916.1 x 323	12	corners A-D from the drawing; rib positions on the outside face
E2	Body front face panel	12 ply	1	630 x 363.3	12	bevels: top 30 deg, bottom 12 deg (see edge profile)
E3	Body floor panel	12 ply	1	773.6 x 630	12	bevels: front 12 deg, rear 18 deg; set-out of E9/E14/F3 on the underside
E4	Tail lip	12 ply	1	630 x 43.9	12	bottom edge bevelled 18 deg
E5	Canopy	12 ply	1	654 x 248.9	12	rear edge bevelled 30 deg
E6	Body top rail	18x44 PAR (1x2)	2	1158	18	flat on the side, flush with its top; 241.9 forward of corner A
E7	Canopy front stiffener	18x44 PAR (1x2)	1	654	18	laid flat under the canopy front edge
E8	Floor-edge rail	18x44 PAR (1x2)	2	776.8	18	two identical pieces (flip one); ends 78 / 72 deg to the long edge
E9	Hinge bracket	18 ply	2	124.1 x 120	18	top cut 18 deg; position on E3 drawing; holes 1 x Ø12.5 (see drawing)
E10	Body side rib 1	12x25 PAR (1/2x1)	2	204.4	12	flat on outside of side, between floor-edge rail and top rail; bottom cut 18 deg
E11	Body side rib 2	12x25 PAR (1/2x1)	2	158	12	flat on outside of side, between floor-edge rail and top rail; bottom cut 18 deg
E12	Body side rib 3	12x25 PAR (1/2x1)	2	111.5	12	flat on outside of side, between floor-edge rail and top rail; bottom cut 18 deg
E13	Body side rib 4	12x25 PAR (1/2x1)	2	65	12	flat on outside of side, between floor-edge rail and top rail; bottom cut 18 deg
E14	Prop pivot block	44x44 PAR (2x2)	1	60	44	under the floor, right face on the centreline; position on E3 drawing; holes 1 x Ø6.5 (see drawing)
F1	Prop arm	18x44 PAR (1x2)	1	319	18	M6 pivot; held folded by the F4 turn-button; holes 1 x Ø6.5 (see drawing)
F3	Prop catch block	44x44 PAR (2x2)	1	44	44	under the floor right of the folded prop; position on E3 drawing
F4	Prop turn-button	12x25 PAR (1/2x1)	1	70	12	Ø5 hole 15 from one end; 4.0x50 screw + washer into F3; holes 1 x Ø5 (see drawing)
G1	Ladder stile	12x25 PAR (1/2x1)	2	310	12	ladder outside faces 330 / 230 left of centre, tops flush with the deck; holes 5 x Ø12 (see drawing)
G3	Ladder fixing block	44x44 PAR (2x2)	1	102	44	under the deck front-left, against B8
H3	Spacer washer 18 mm	18 ply	8	Ø60	18	4 front (inboard of wheel), 4 between rear duals; holes 1 x Ø20 (see drawing)
H4	Spacer washer 12 mm	12 ply	4	Ø60	12	between hinge post and inner rear wheel; holes 1 x Ø20 (see drawing)
H1	Front axle	20 mm steel bar	1	642	Ø20	held by collars (does not turn; wheels turn on it); or drill 4 mm split-pin holes 17 mm in from each end
H2	Rear axle	20 mm steel bar	1	700	Ø20	held by collars (does not turn; wheels turn on it)
F2	Body hinge pin (stub)	12 mm dowel	2	42	Ø12	glue into hinge post only (12 mm hole); body bracket hole 12.5 so it swings
G2	Ladder rung	12 mm dowel	5	100	Ø12	glue into stile holes
W	Wheel 4.00-6 PU 330x100 (bought)	purchased	6	Ø330	100	Wheelbarrow Supplies, £15.95 each

ANGLED CUTS (all other cuts square; angles from the part drawings)

- E1 Body side (wedge) x2: front edge 60 deg to the top edge (corner A); floor edge 18 deg to the top edge; keel corner B 102 deg, tail corner C 108 deg; tail edge square (D 90).
- C1 Cab side x2: back edge 60 deg to the bottom edge (top-back corner 120 deg) - parallel to the body front.
- E8 Floor-edge rail x2 (two identical pieces, flip one): front end 78 deg to the long edge (12 off square), rear end 72 deg (18 off square).
- E10-E13 Ribs x8: bottom end 18 deg off square. E9 Hinge bracket x2 and A6 Rest block x2: top edge 18 deg to the bottom edge.
- EDGE BEVELS - saw blade tilted, degrees off square; required, they make the panel joints meet (long point shown on each part's edge profile): C3 both ends 30 (parallel); C4 rear 30; E2 top 30, bottom 12; E3 front 12, rear 18; E4 bottom 18; E5 rear 30.
- Mitre/jigsaw settings: a 60 deg corner = 30 deg off square; 108 deg = 18 deg off square. Everything else is cut square.

Straight through-cut (guillotine) layouts: every cut runs right across the piece being cut (track saw / table saw). 4 mm kerf, 8 mm edge trim. Long parts along the sheet length (grain). Discs, wedges, bevelled panels = blank rectangle.

TIMBER / DOWEL / STEEL TO BUY

(5 mm allowed per saw cut. Where the spare is under 100 mm, buy straight, defect-free stock or one extra length.)

18x94 PAR (1x4 in): 2 x 2400 mm lengths (parts total 3260 mm)

- length 1: A1 1062, A1 1062, A2 164 (spare 97)
- length 2: A3 480, A2 164, A2 164, A2 164 (spare 1408)

44x44 PAR (2x2 in): 1 x 2400 mm length (parts total 758 mm)

- length 1: G3 102, A4 94, A4 94, A4 94, A4 94, E14 60, B9 44, B9 44, B9 44, B9 44, F3 44 (spare 1587)

18x44 PAR (1x2 in): 3 x 2400 mm lengths (parts total 5682.6 mm)

- length 1: E6 1158, E6 1158 (spare 74)
- length 2: E8 776.8, E8 776.8, E7 654 (spare 177.4)
- length 3: B8 420, B8 420, F1 319 (spare 1226)

12x25 PAR (1/2x1 in): 1 x 2400 mm length (parts total 1768 mm)

- length 1: G1 310, G1 310, E10 204.4, E10 204.4, E11 158, E11 158, E12 111.5, E12 111.5, F4 70, E13 65, E13 65 (spare 577)

12 mm hardwood dowel: 1 x 1200 mm length (parts total 584 mm)

- length 1: G2 100, G2 100, G2 100, G2 100, G2 100, F2 42, F2 42 (spare 581)

20 mm bright mild steel round bar: 2 x 1000 mm lengths (parts total 1342 mm)

- length 1: H2 700 (spare 295)
- length 2: H1 642 (spare 353)

SHEETS

- 12 mm plywood: 1 x 2440 x 1220
- 18 mm plywood: 1 x 1220 x 607
- 6 mm plywood: 1 x 400 x 200 (offcut; or buy a 1220 x 607 sheet)

HARDWARE

- 6 x wheels 4.00-6 PU 330x100, 20 mm bushes (see sheet 12)
- 4 x 20 mm shaft collars (grub screw) + 4 x M20 flat washers (or drill axle ends 4 mm for split pins, 17 mm from each end)
- 1 x M6 x 80 bolt, 2 washers, nyloc nut (prop pivot)
- Hinge: 2 x 12 mm dowel stub pins (cut from the dowel, see F2)
- Wood screws 4.0 mm countersunk, pilot-drilled (buy a few spare):

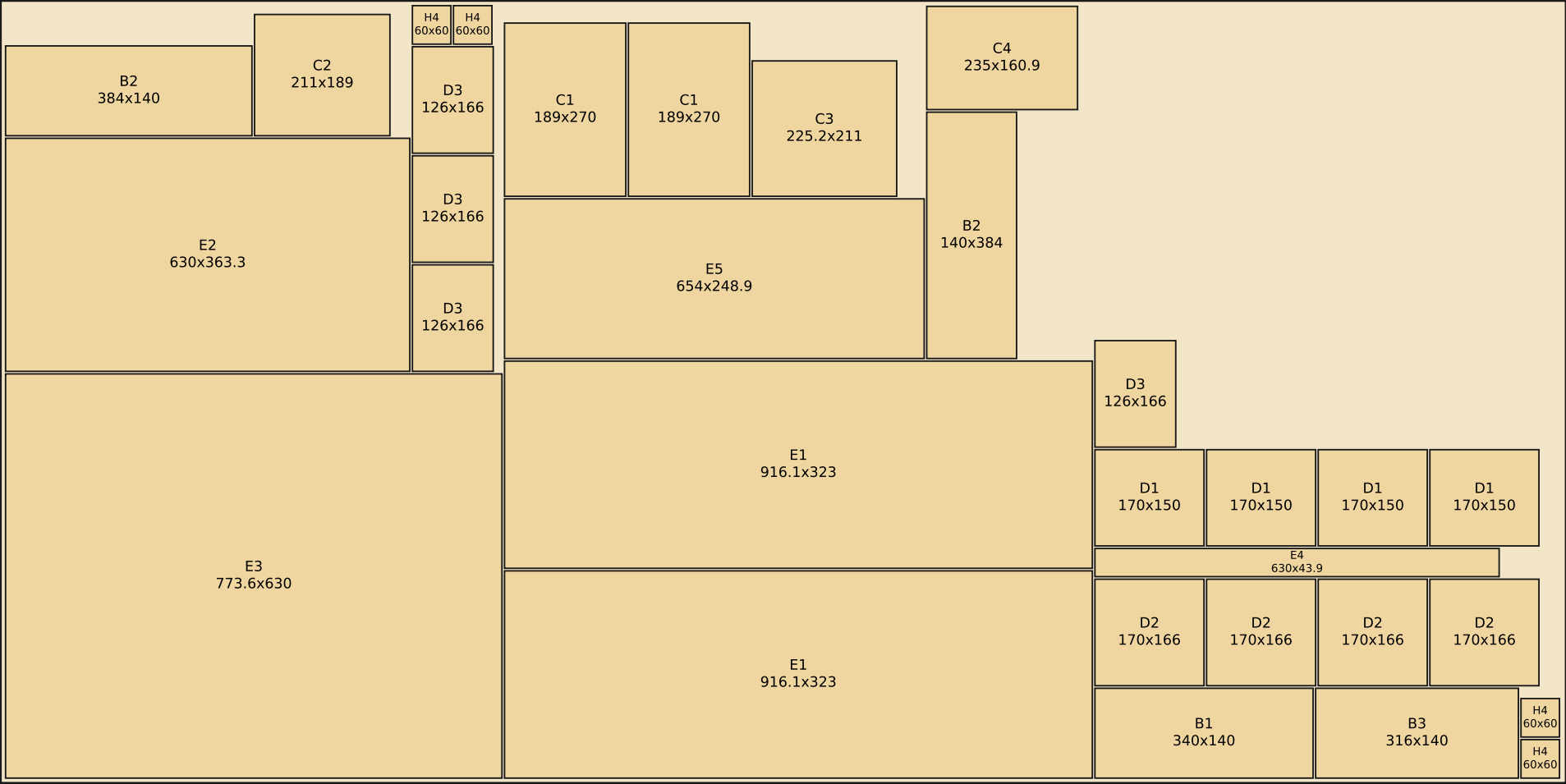
25 mm x 60: 12 mm ply joints, E6/E8/E7 rails, tanks

40 mm x 55: 18 mm parts (rails, bumper, posts, deck, E9, ladder)

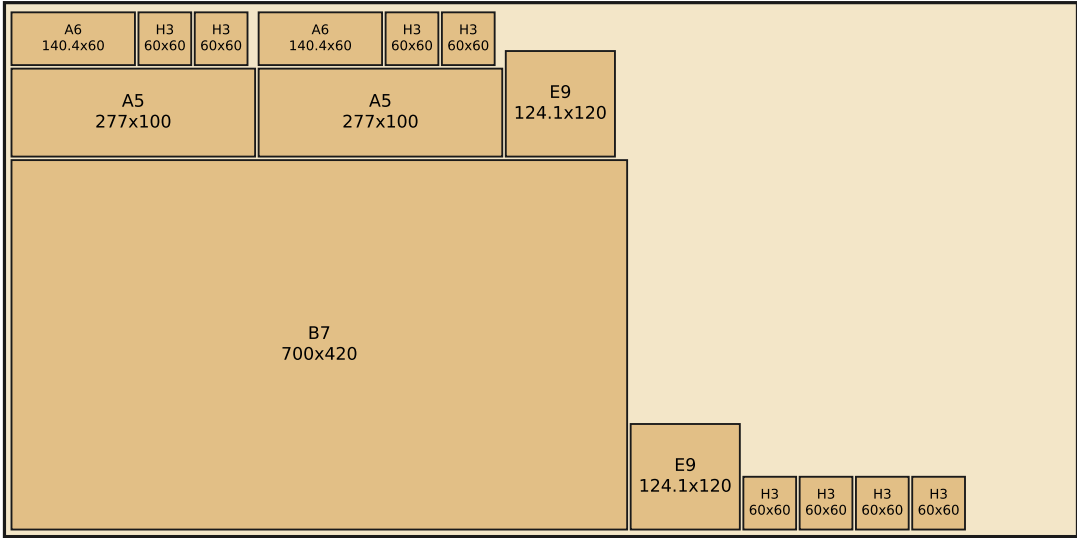
50 mm x 12: into the 44 mm blocks (A4, E14, F3, G3) + 1 for F4

70 mm x 4: B9 engine-box blocks down into the rails
- 1 x 4 mm washer (F4 turn-button)
- 25 mm panel pins / brad nails, PVA wood glue
- Primer + paint (CAT-style yellow, black, grey)

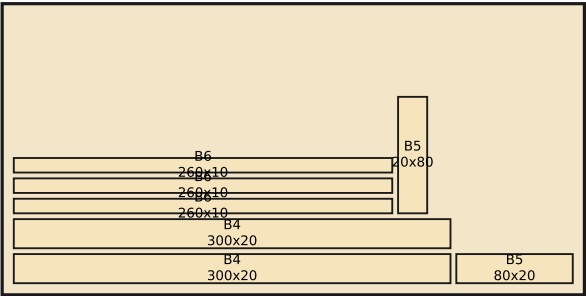
12 mm plywood sheet 1 (2440 x 1220)



18 mm plywood sheet 1 (1220 x 607)



6 mm plywood sheet 1 (400 x 200)



ASSEMBLY ORDER (glue = PVA; screws pilot-drilled; pins 25 mm)

1 Prepare

Cut all parts from the cutting list (angled cuts and edge bevels are listed on the cutting-list sheet). Mark L/R pairs. Drill the 20 mm axle holes in the A1 rails, A4 blocks and A5 hinge posts with the parts clamped together so the holes line up (drill press best). Drill the 12 mm hinge-pin holes in both A5 posts and the 12.5 mm holes in both E9 brackets as matched pairs. Mark the positions shown dashed on the A1, B1, B7, E1 and E3 drawings. Sand edges before assembly.

2 Chassis

Lay the two A1 rails on edge on a flat bench, outer faces 200 apart, front ends level. Glue + screw (4.0x40, through the rails) the four A2 cross members between them, flush top and bottom, front faces 0, 462, 652 and 1044 from the rail FRONT end (= x 18, 480, 670 and 1062 from the bumper front; the 3rd one is the prop seat). Check square by measuring diagonals. Glue + screw the A3 bumper across the front ends: it also glues to the face of the 1st cross member.

3 Axle mounts + hinge

Glue the A4 blocks together in pairs, then glue + screw (4.0x50 through the rail) a pair to the outside of each rail at the front axle, holes lined up (push a 20 mm rod through while the glue sets). Glue + screw the tall A5 hinge posts to the outside of the rails, centred on the rear axle (rod through the axle holes while the glue sets): the hinge pin hole is then exactly 200 mm above the rear axle. Glue the A6 rest blocks on top of the rails, front end 482 from the rail front end (x 500), slope rising to the rear.

4 Front end

Build the radiator/engine box (B1 front, B2 sides, B3 back) with glue + 25 mm pins, and glue the four B9 blocks inside it in the bottom corners (behind B1 and in front of B3, positioned to sit on the rails). Stand the box on the rails 24 mm back from the bumper front and screw down through each B9 block into the rail top (4.0x70), 9 mm in from the block's inboard face so the screw enters the middle of the rail. Glue the 6 mm grille frame (B4, B5) and slats (B6) on the front, 20 in from each side and 10 up from the bottom. Glue + screw the B7 deck on top, flush with the bumper front, then the B8 edge rails under its long edges and the G3 ladder block under its front-left corner against the left B8 rail.

5 Cab, tanks, ladder

Glue the cab front C2 between the two trapezoid C1 sides, then the sloping back C3 (both ends bevelled 30 deg) between them with its outer face flush with their 60 deg back edges, then the roof C4 (back edge bevelled 30 deg). Paint the windows. Screw the cab to the deck from below on the LEFT side, 15 mm in from the deck edge, with the foot of the sloping back exactly on the deck's rear edge (x = 420). Screw each tank inner side (D1) to the rail outer face, front edge 452 from the rail front end and bottom edge 32 above the rail underside, then glue the rest of the box onto it. Glue the dowel rungs into the ladder stiles; screw each stile (4.0x40) to the deck front edge and the G3 block, the ladder's two outside faces 330 and 230 left of the centreline, tops flush with the deck.

6 Body

Lay one E1 wedge side flat. Glue + screw the floor E3 to it (underside flush with the 18 deg bottom edge, bevelled front edge on the 60 deg line, rear edge flush with the tail), then the front face panel E2 (outer face flush with the 60 deg front edge, top level with the side top, bevelled bottom edge sitting on the floor), then the tail lip E4 on the floor, flush with the tail. Add the second side. Glue + screw the E6 top rails along the top outside edges, running 241.9 mm forward of the sides; fit the canopy E5 between their front ends, its bevelled rear edge on the front face, and the flat E7 stiffener under its front edge. Add the E8 floor-edge rails and the ribs E10-E13 (glue + pins). Turn the body over, mark the E3 underside set-out from the keel and drill small pilot holes through the floor inside each outline: the E9 hinge brackets (front end 340.2 from the keel, outer faces 100 off centre), the prop block E14 (front end 136.3, right face on the centreline) and the prop catch block F3 (front end 400, 2 mm right of where the prop folds). Glue the parts on, then turn the body upright and screw down through the pilot holes into them (4.0x40 into E9; 4.0x50 into E14 and F3).

7 Hinge + prop

Sit the body on the rest blocks with the E9 brackets just inside the A5 posts. Tap a 12 mm dowel stub pin F2 through each post into its bracket, glued into the post only so the body can swing. Check the 10 mm gap between the cab back and the body front is even. Bolt the prop F1 to the right side of its block with the M6 bolt: it folds back under the floor, where the F4 turn-button (screwed to the underside of F3 with one 4.0x50 screw and a washer) turns across under it to hold it. Keep the button across whenever the body moves. To prop the body: lift it to 45 deg by hand, turn the button along the floor, swing the prop down and stand its foot on the 3rd cross member. To lower: lift slightly, fold the prop up under the floor and turn the button back across it, then lower the body.

8 Wheels

Paint before fitting wheels. Push the 20 mm steel axles through (they are held by the collars and do not turn; the wheels turn on them). Front each side: 2 x 18 mm ply spacers, wheel, washer, collar. Rear each side: 2 x 12 mm spacers, inner wheel, 2 x 18 mm spacers, outer wheel, washer, collar. Fit the wheel's 20 mm reducer bushes first. Measure your real hub length (design assumes 75 mm) and adjust spacer thickness if needed.



Reference: Caterpillar 777D (photo: Jo Mateix, public domain, Wikimedia Commons)
https://commons.wikimedia.org/wiki/File:Caterpillar_777D.jpg

WHEEL - design basis

Wheelbarrow Supplies: 12"/13" Wheel 4.00-6 PU Puncture Proof Solid Wheel - 20mm Bore - 13 x 4

330 mm dia x 100 mm wide, 6" plastic rim, 35 mm bore + bushes to 20 mm, hub approx 75 mm

£15.95 each (free UK mainland delivery shown). Need 6 = £95.70

wheelbarrowsupplies.com/12-wheel-4-00-6-pu-puncture-proof-solid-wheel-16mm-bore-...-1.html

(ignore '16mm' in the web address: the page says 35 mm bore + bushes to 20 mm)

ALTERNATIVES

B&Q (sold by Keto Plastics): 14" 3.50-8 puncture-proof, 350 x 90, 25 mm bore, hub 75, £15.95

-> 20 mm bigger: raise axle_z by 10 (all rails/deck follow); use 25 mm axle

Ollies Trolleys: 13" x 5" foam-filled 330 x 125, 16 mm bore, hub 58, £19.99

-> chunkier 'mining' look; change wheel_w/hub_len/axle_d in design.py and rebuild

TIPS

- Paint parts before final assembly where possible (wheels off!).
- Weight approx 35-40 kg (wood ~24, steel ~3, wheels ~2 each): two-person lift, by the chassis.
- Dry-fit the body on the hinge and check the cab gap before gluing the cab down.
- Angled cuts and edge bevels: see the list on the cutting-list sheet (60, 18, 30 and 12 deg).
- G1 stiles: 12 mm holes leave 6.5 mm each side - drill slowly with a backing board.